Engineering Portfolio

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Coursework Projects

My coursework projects integrate control theory, dynamics, and vibration analysis to model, design, and optimize real-world electrical and mechanical systems. Through these projects, I’ve developed experience in applying feedback control, system identification, and simulation-based design to achieve stable and efficient system behavior.

The drone controller project allowed me to tune and implement PID control loops, address nonlinear effects such as actuator saturation and integrator windup, and refine performance using frequency-domain methods. Projects in dynamic system optimization and vibration isolation further strengthened my understanding of how physical parameters and control strategies interact to govern system response.

Together, these projects demonstrate a strong foundation in system optimization and the ability to connect analytical models with experimental performance.

Project: Quadcopter Drone Simulation

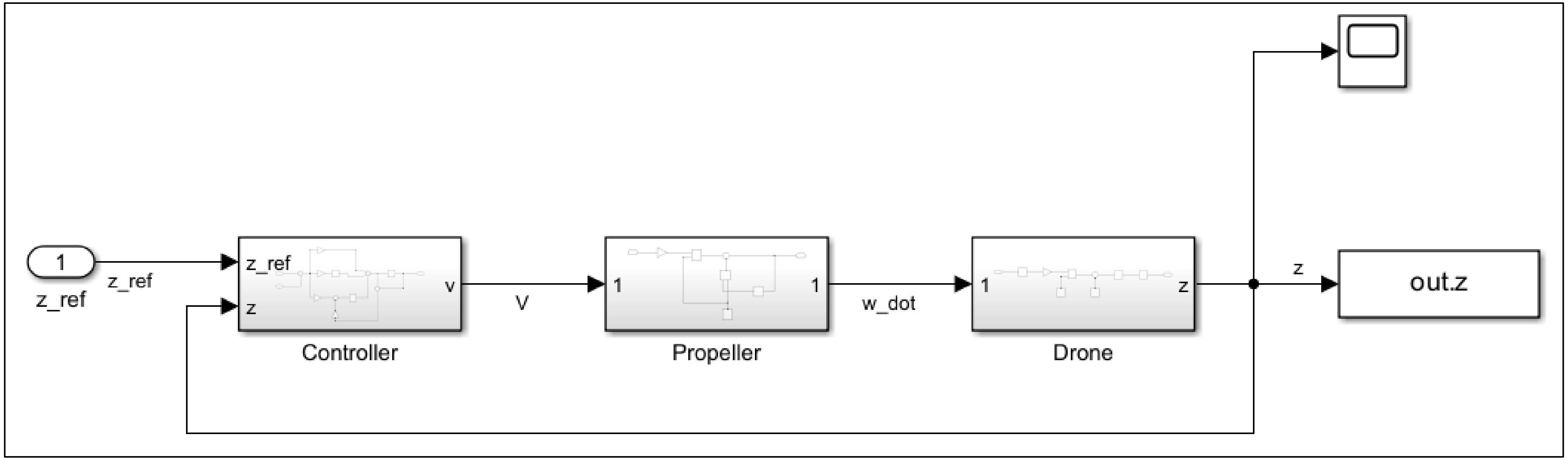
For the final project in our Control Systems Engineering course, my partner and I developed a quadcopter drone controller from scratch. We first derived a mathematical model of the quadcopter dynamics using Newton-Euler equations, capturing the translational and rotational motion of the drone. We then linearized the system about a hover point and designed a cascade of PID controllers for position control.

We implemented a motor mixing scheme to map the desired total thrust, and body torques into individual rotor commands. We tuned the PID gains to achieve stable hover, fast transient response, and minimal overshoot in response to reference inputs and a wind simulation.

We implemented and tested the drone in MATLAB/Simulink. We created a three-dimensional waypoint navigation scenario to evaluate position control as the drone tracked commanded trajectories through space. The simulation incorporated realistic dynamics and disturbances such as gravity and air resistance, allowing us to accurately assess controller performance.

Controller Design and Stability

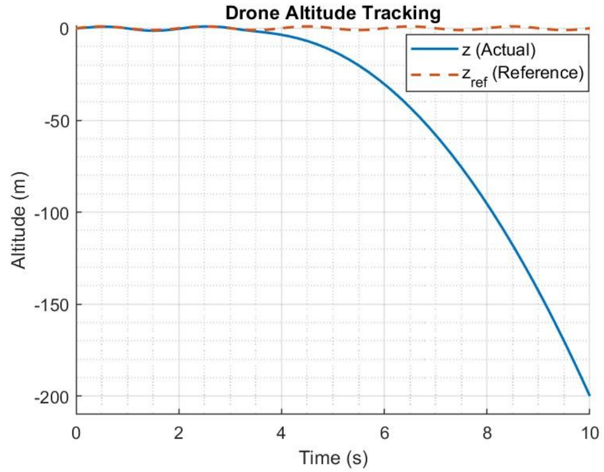
- Control loop implemented using multiple PID controllers that regulate each propeller's thrust to achieve a desired position
- Nonlinearities in roll/pitch angles and propeller voltage saturation affect stability but can be linearized about stable hover



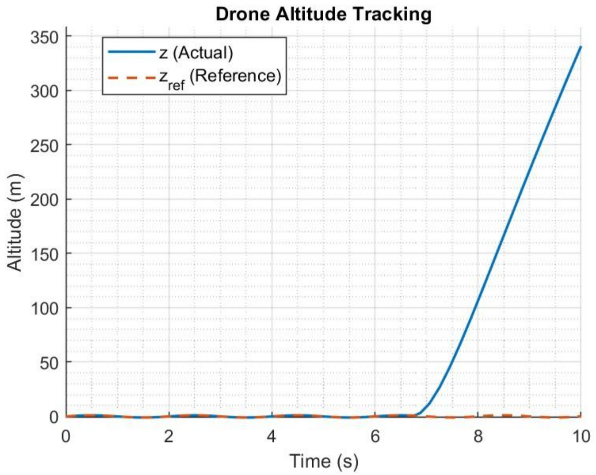
Tracking & Integration Windup

- Case 1 shows a drone whose propellers are too weak and cannot recover from a slight disturbance like one caused by windup
- Case 2 shows adequately specified propellers still in the windup case. Once the integration term reaches a threshold, the altitude asymptotically rises in this case
- We implemented an anti-windup feedback loop to limit the integrator output when the actuator saturates, preventing error accumulation

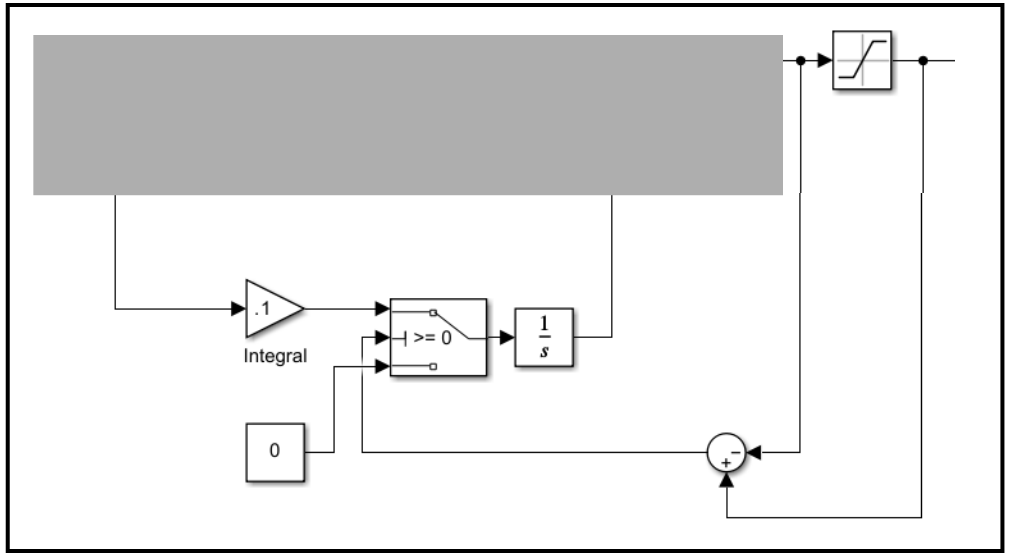
Case 1



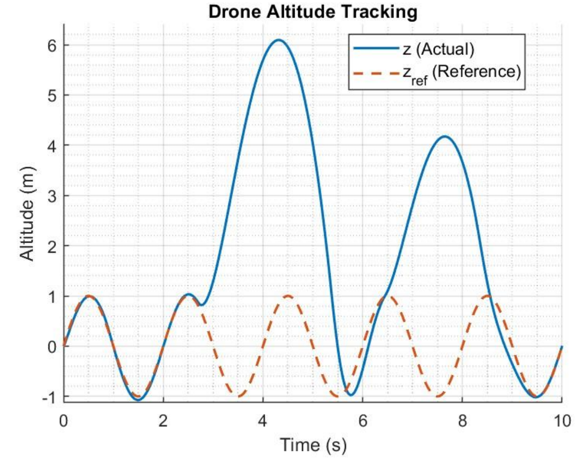
Case 2



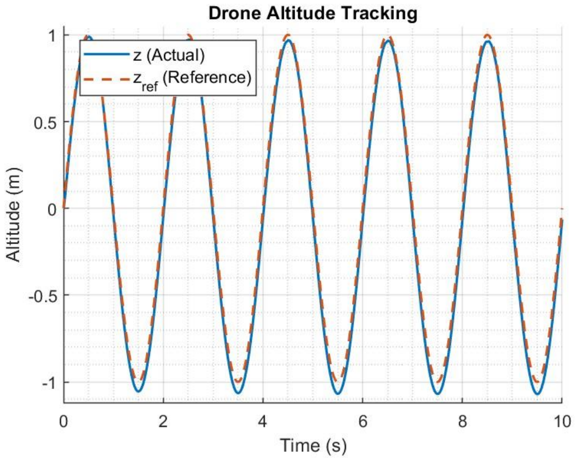
Anti-Windup Feedback Loop



Poor Tracking



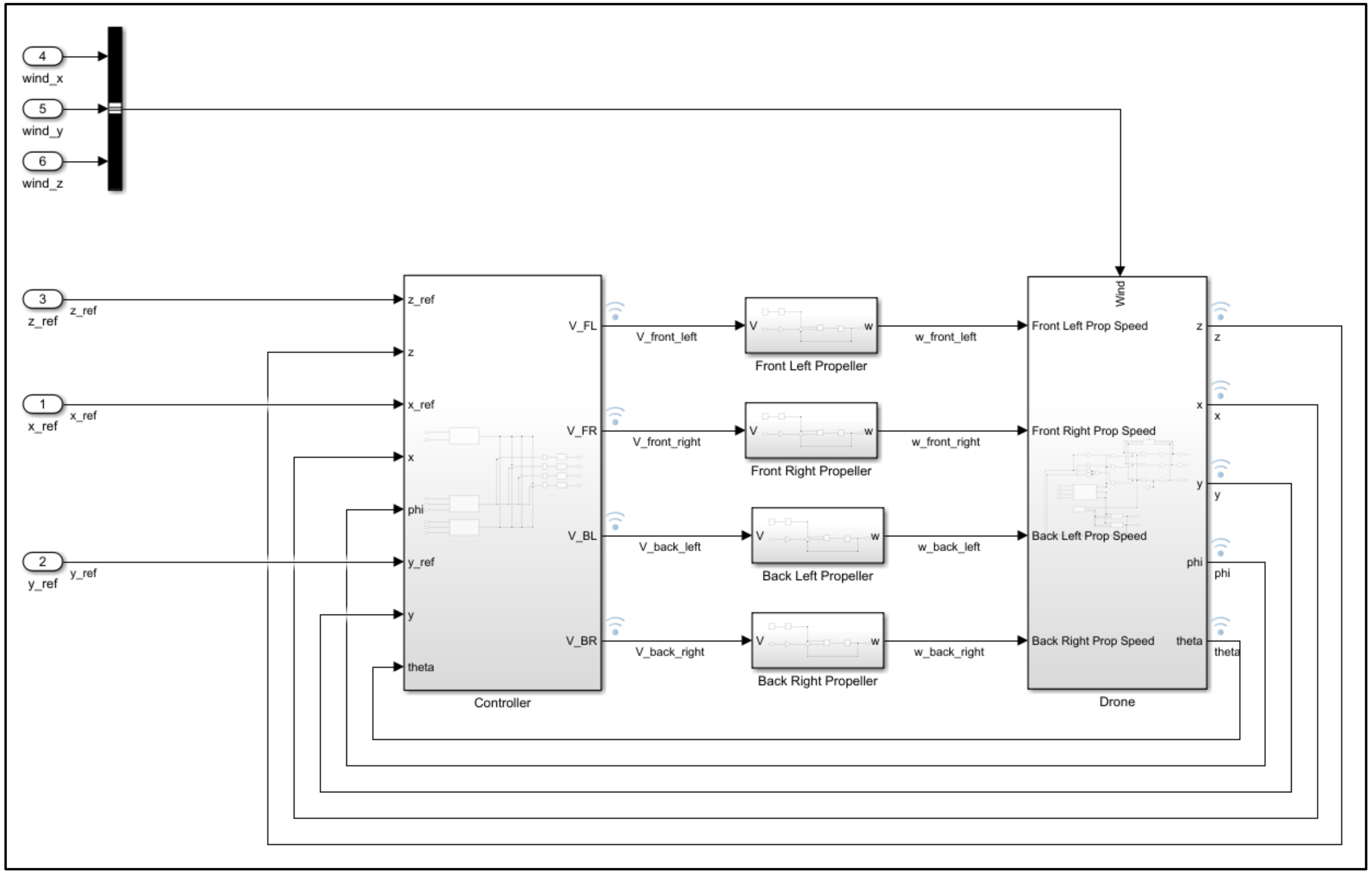
High-Fidelity Tracking



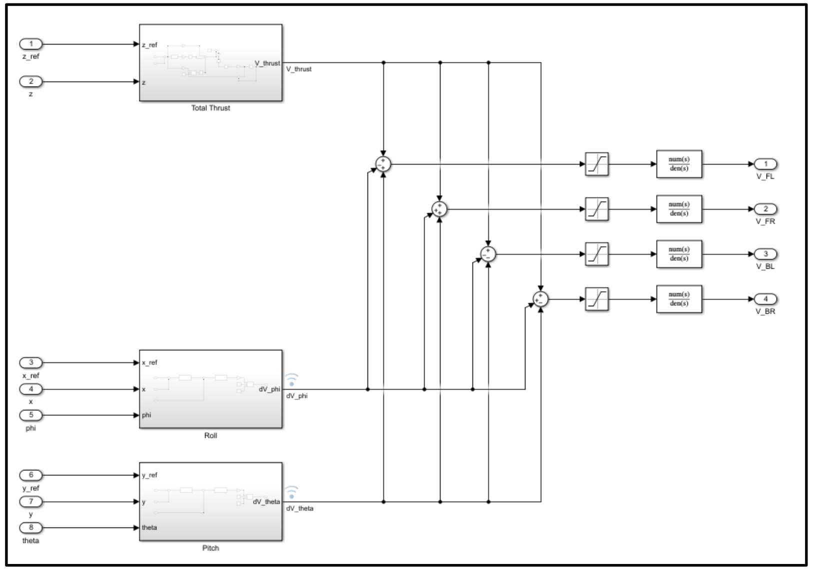
- The Poor Tracking plot shows a stable response but with remaining windup error
- The last plot shows the response after tuning. It displays a responsive system and low steady-state error. Error at the extrema of the reference wave was acceptable by the project requirements

Simulink Block Diagram

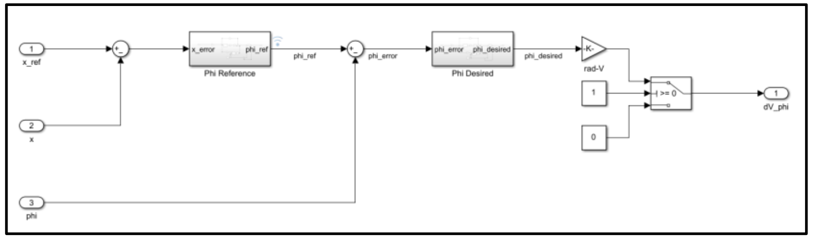
Highest Level: Position Reference Input to Drone Position Output



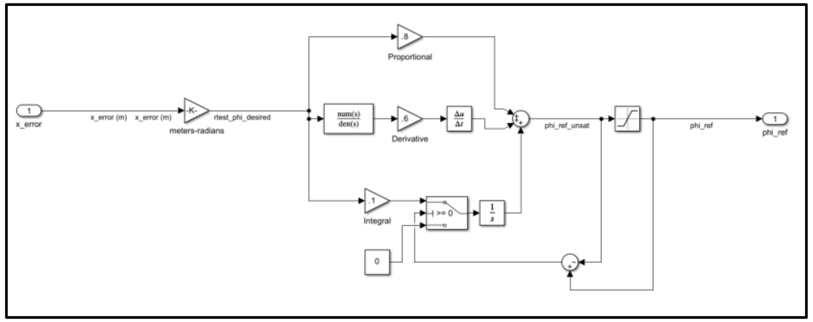
Controller: References to Voltages



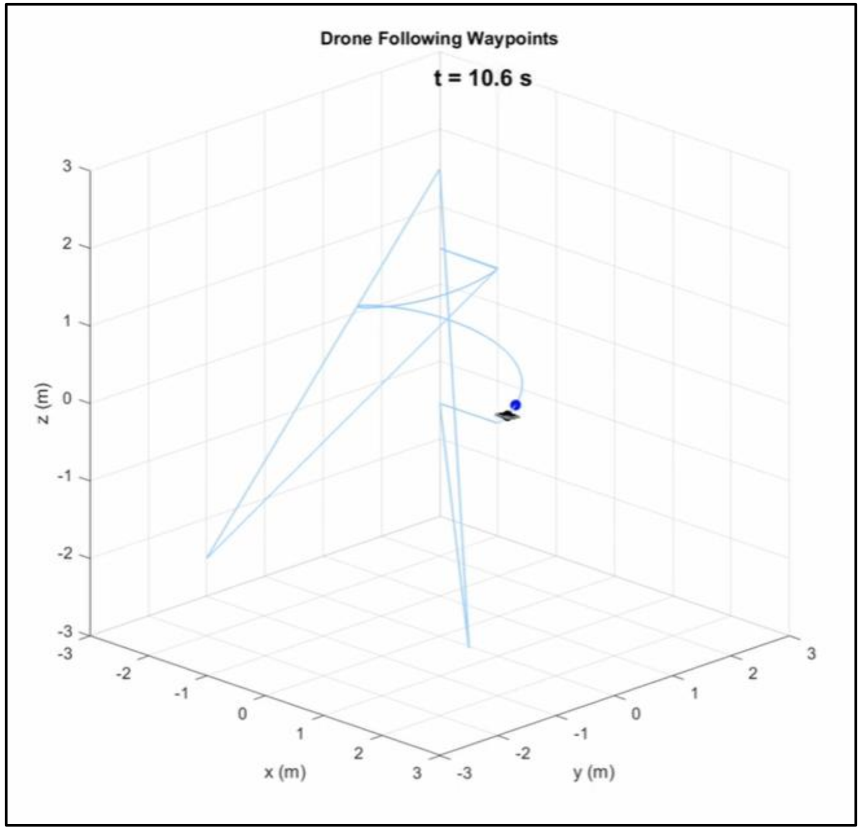
Roll Controller: Reference to ΔVoltage



Roll Angle Controller: Error to Reference



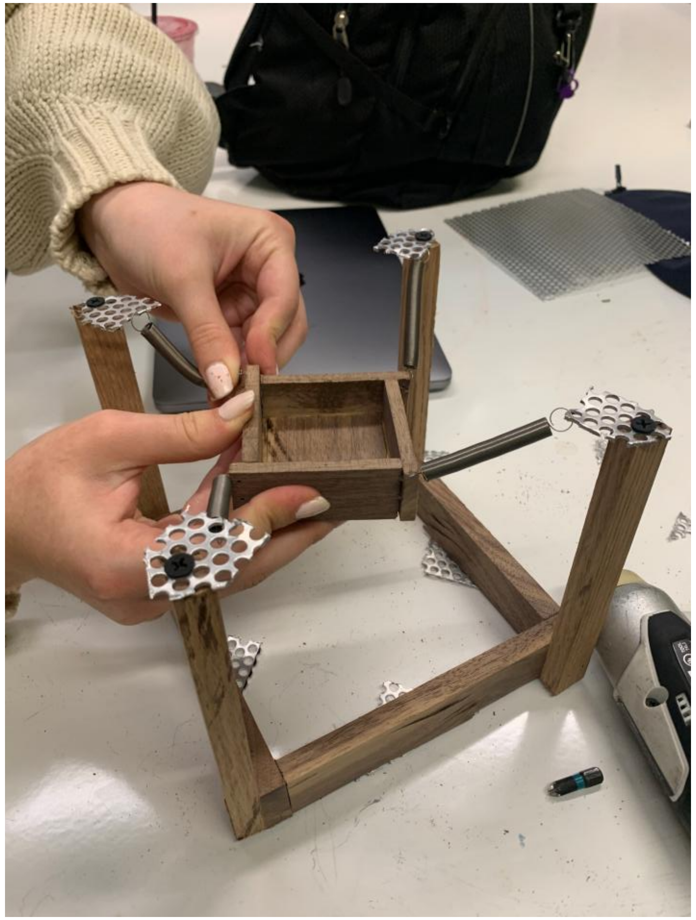
3D Waypoint Simulation



https://www.youtube.com/watch?v=MtzlHR9d-20

Project: Vibration Isolation System

- For a Dynamics and Vibrations project, our team designed and tested a spring-mass vibration system to minimize transmission of shake-table vibrations to a suspended platform
- Used analytical modeling to derive the system's natural frequency (2.1 Hz) and transmissibility (0.0022 at 45 Hz) under sinusoidal excitation
- Experimental natural frequency 2.2Hz, and higher transmissibility due to damping

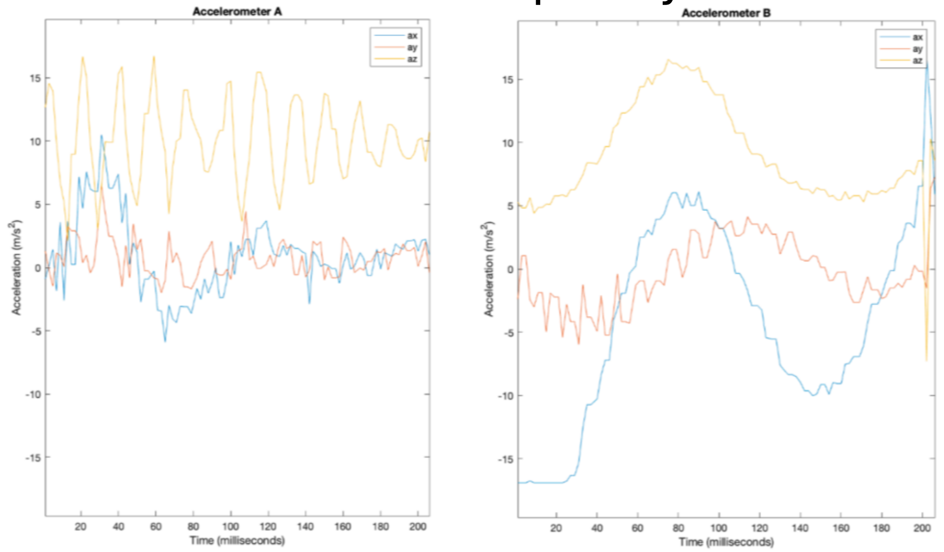


Accelerometer A: shake-table

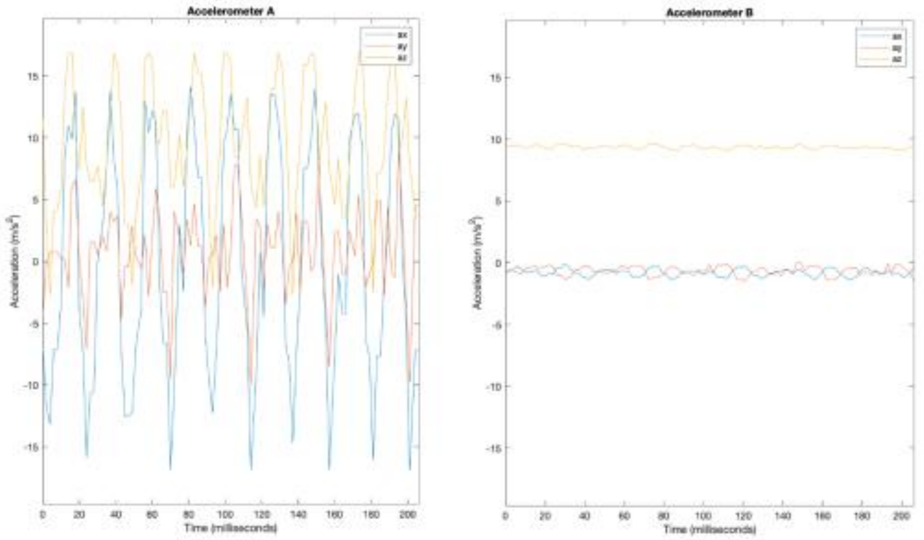
Accelerometer B: platform

- At 2.2 Hz, system resonance caused high amplitude oscillations to throw the mass off the platform
- This then lowered the mass of the platform, raising the natural frequency
- At higher frequencies including 45 Hz, the vibrations were almost completely isolated to the shake-table

Shake-Table Frequency 2.2Hz



Shake-Table Frequency 45Hz

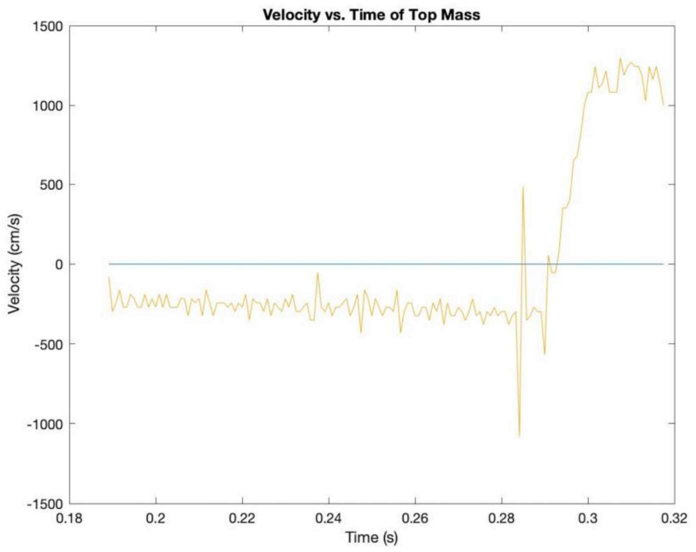


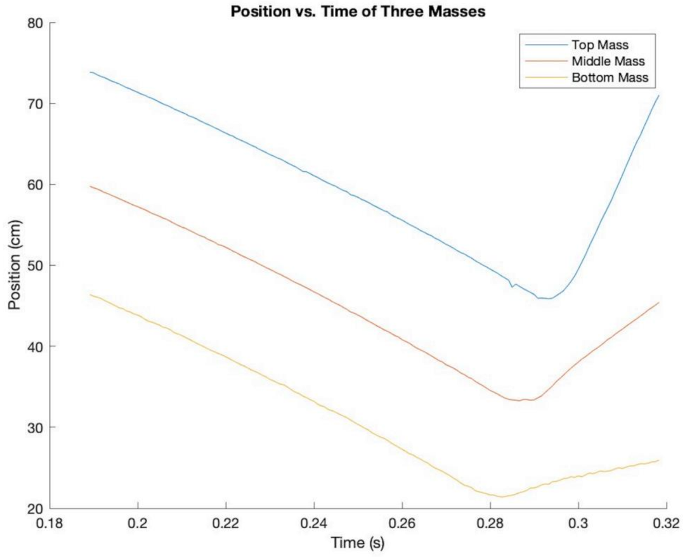
Project: Mass Launcher

- For a Dynamics and Vibrations, my group optimized a three-mass, three-spring projectile launcher for launch velocity
- Used Hooke's Law and Newton's 2nd Law as well as MATLAB's fmincon optimization tool to determine the optimal set of masses and spring constants

Mass Launcher

- Used cameras to track and plot mass positions, then differentiated for velocity
- Experimental launch velocity of 40.7 ft/s closely matched theoretical 43.3 ft/s, validating the analytical model



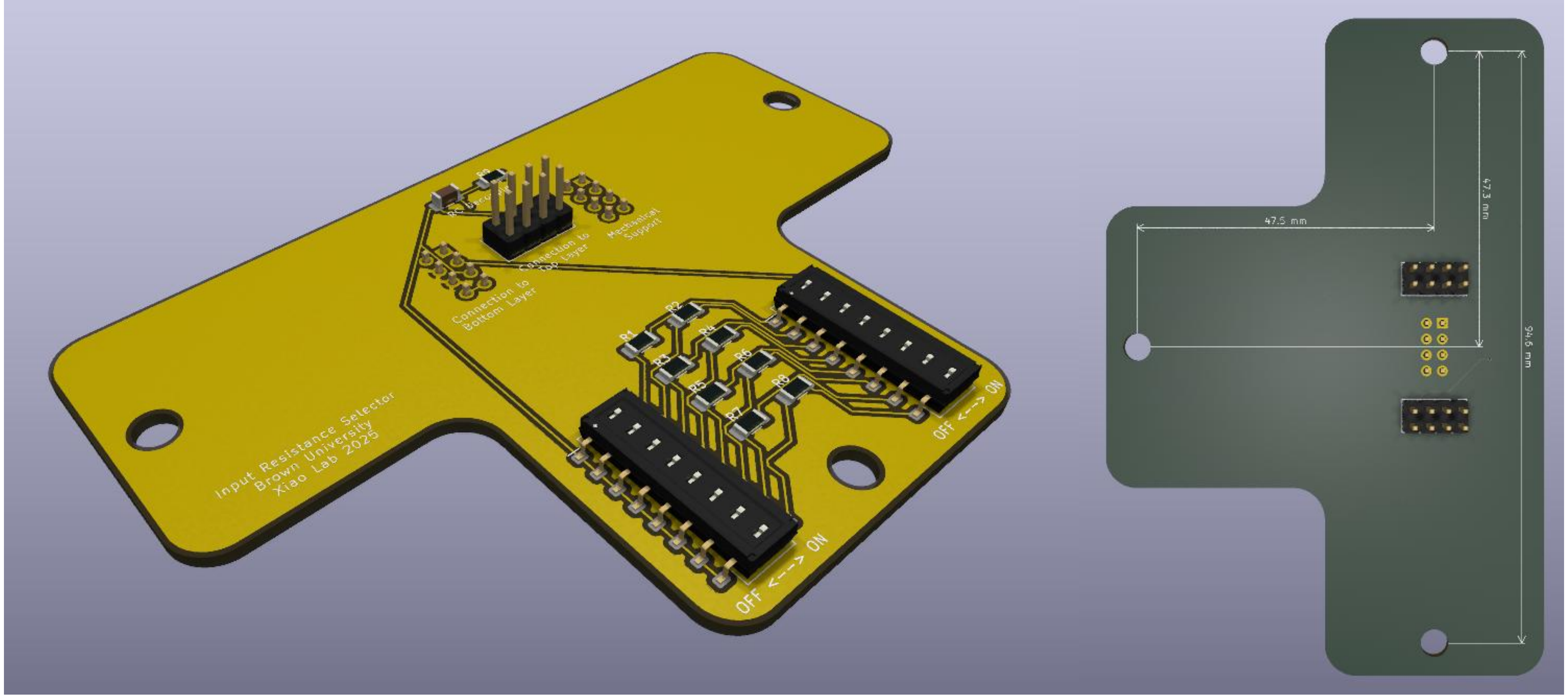


Lab Projects

As an undergraduate research assistant in the Nanoscale Physics and Quantum Devices Lab at Brown University, I work on projects aimed at developing highly sensitive magnetic field detectors capable of measuring fields in the picotesla range with minimal noise. A key focus of the group is understanding the causes of and mitigating electrical noise in magnetic tunnel junction (MTJ) devices.

My work specifically centers on characterizing and reducing intrinsic noise in MTJ arrays. Working closely with the member of the team responsible for running measurements in the Faraday cage, I help design, test, and analyze systems that sample intrinsic electrical noise. These measurements are performed by connecting the MTJ array to a DC voltage source and data acquisition system (DAQ) without applying an external magnetic field, then analyzing the resulting noise spectra in the frequency domain.

Project: Input Resistance Selector Circuit



Purpose

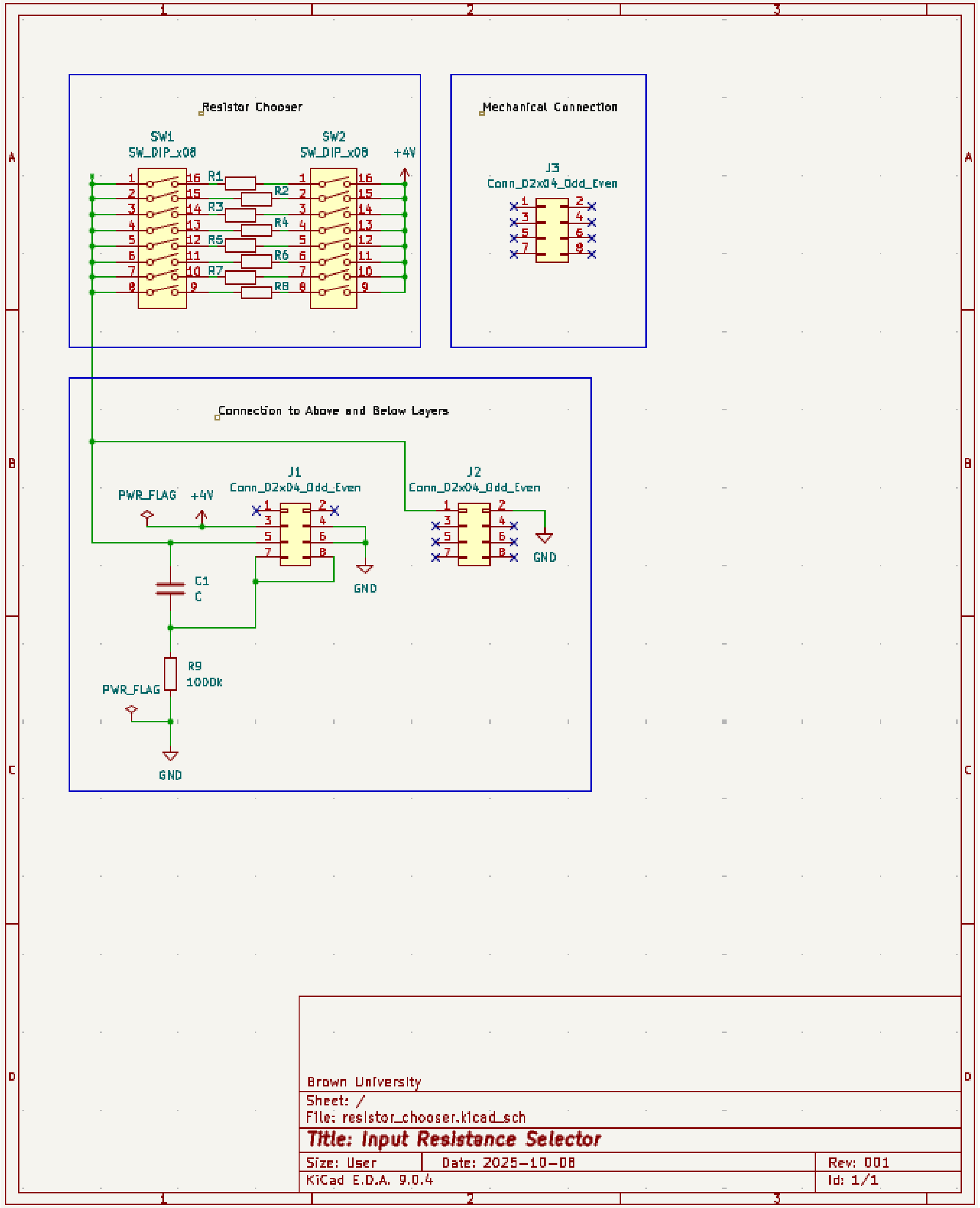
The first project I designed is a selectable input resistance circuit that allows us to control the effective bias voltage applied across an MTJ array during noise measurements. By selecting different input resistances, we can tune the bias conditions experienced by the MTJs, enabling exploration of how bias voltage affects their intrinsic noise characteristics.

Because even small amounts of additional electrical noise can compromise the sensitivity of the measurement, the circuit must introduce minimal noise of its own. Careful component selection, grounding, and shielding were therefore essential in its design. One main feature of the circuit is to electrically isolate unused resistors that would otherwise overwhelm the intrinsic noise measurement. The circuit is part of a larger multi-stage assembly and must integrate seamlessly with the subsystems above and below it:

- The upper stage connects to the MTJ array.
- The lower stage interfaces with the DAQ system and battery-powered voltage source, which includes low-pass filters and amplifiers for signal conditioning.

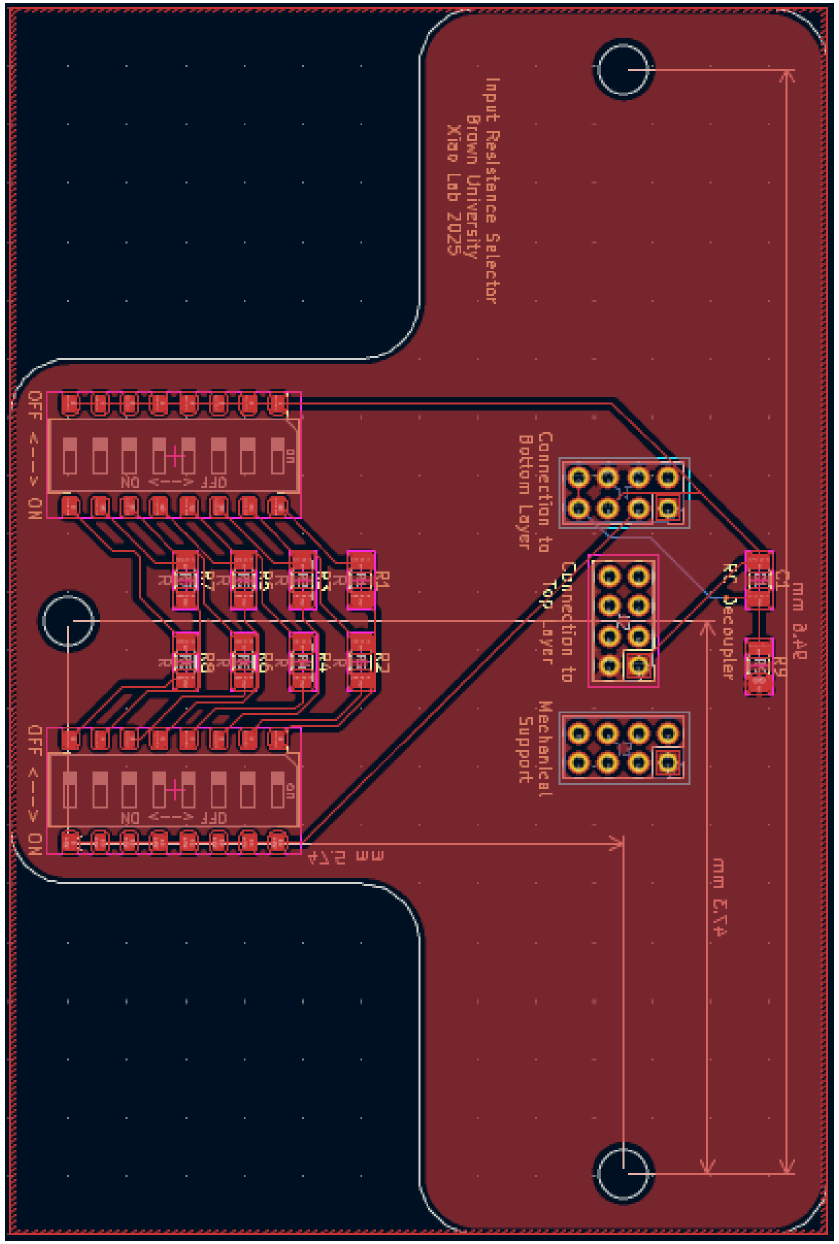
Design & Schematic

- 2 DIP switches to choose between discrete input bias voltages
- Electrical and mechanical connections to other stages
- Isolates thermal noise caused by resistors carrying no current
- Features RC rectifier stage to suppress low-frequency components in data acquisition



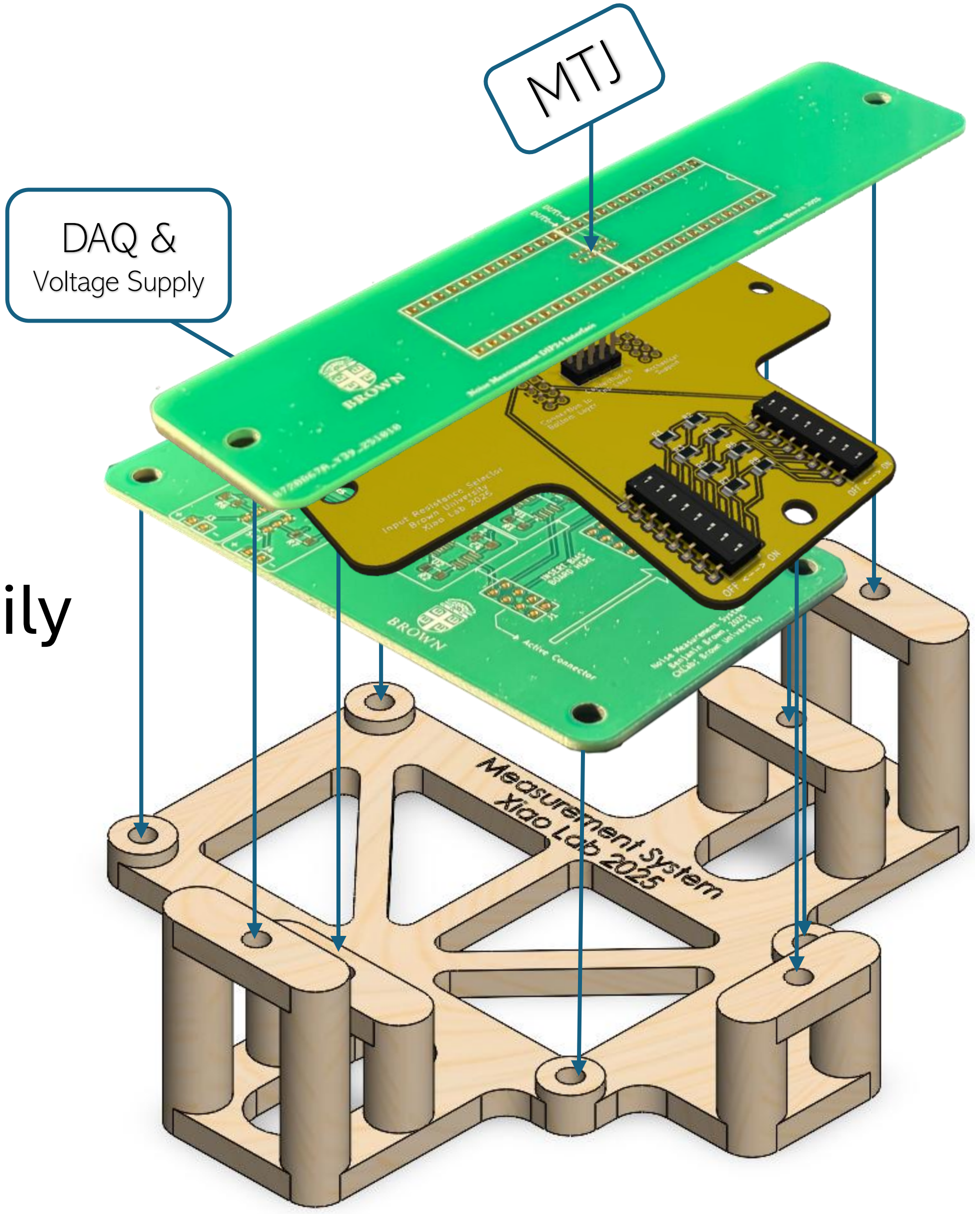
Circuit Board Design

- Short trace lengths to reduce parasitic capacitance and inductance
- Ground plane provides a low-impedance return path for signal from MTJ array
- Components and dimensions well marked for easy assembly and use



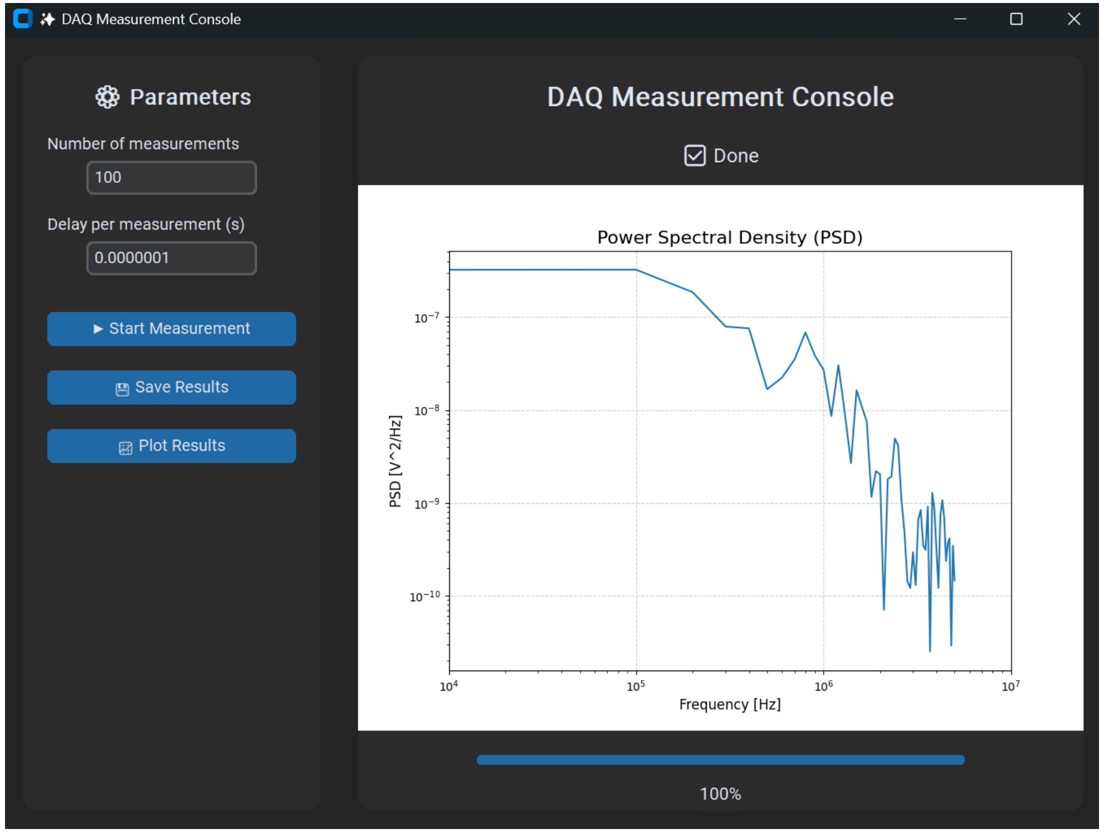
PCB Assembly

- DIP switches accessible without removing top layer board
- Designed a 3D printed mount to support PCBs so top layer can be easily swapped between measurements



Project: Measurement User Interface

- Making an interface to enable real time monitoring of measurements that can take hours/days to finish
- Once completed, GUI will upload live plots as data is collected for monitoring from outside Faraday cage and outside lab



FSAE Projects

I joined Brown’s FSAE team in early 2024, machining random parts for different subsystems. Sophomore year, I started designing my own parts for the Suspension and Pedalbox, and as a Junior I am now the Lead Responsible Engineer for the Drivetrain subsystem. My design work on the drivetrain entails running dynamic simulations and structural analyses on components.

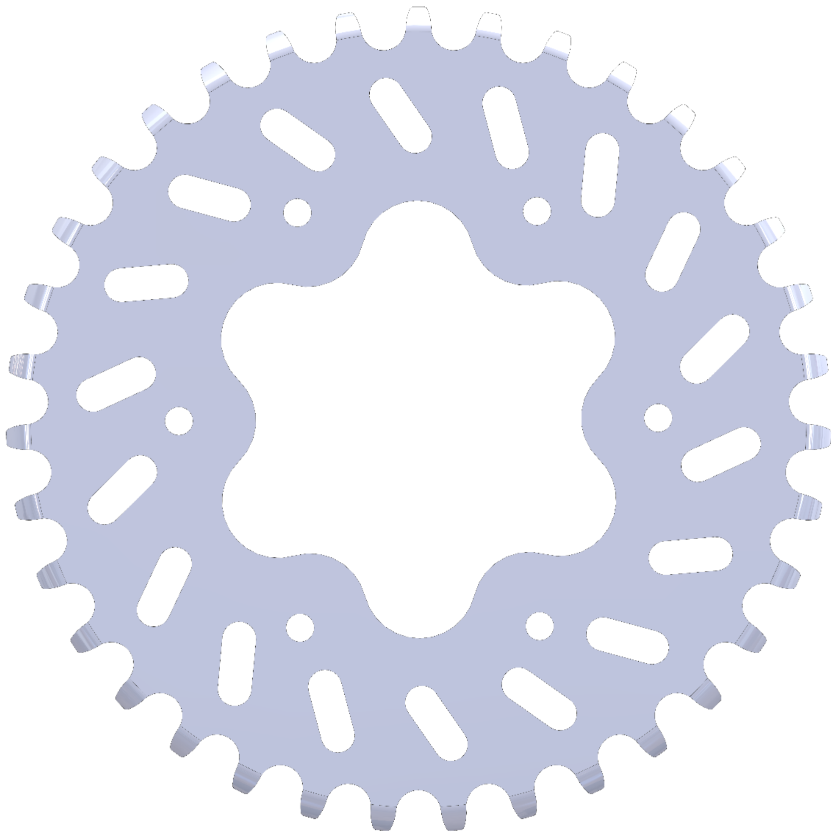
For purely acceleration simulations that do not require differential tuning, I write MATLAB scripts to compare vehicle parameter configurations. For simulations with turns, I have started using a lap sim software, where I can also tune the rear differential.

For structural analysis, I perform force calculations for all the components in the subsystem. I import my CAD drawings into ANSYS, where I perform static load case calculations.

Through January 2025, my subsystem team and I will machine and order different parts so that assembly can begin in late January.

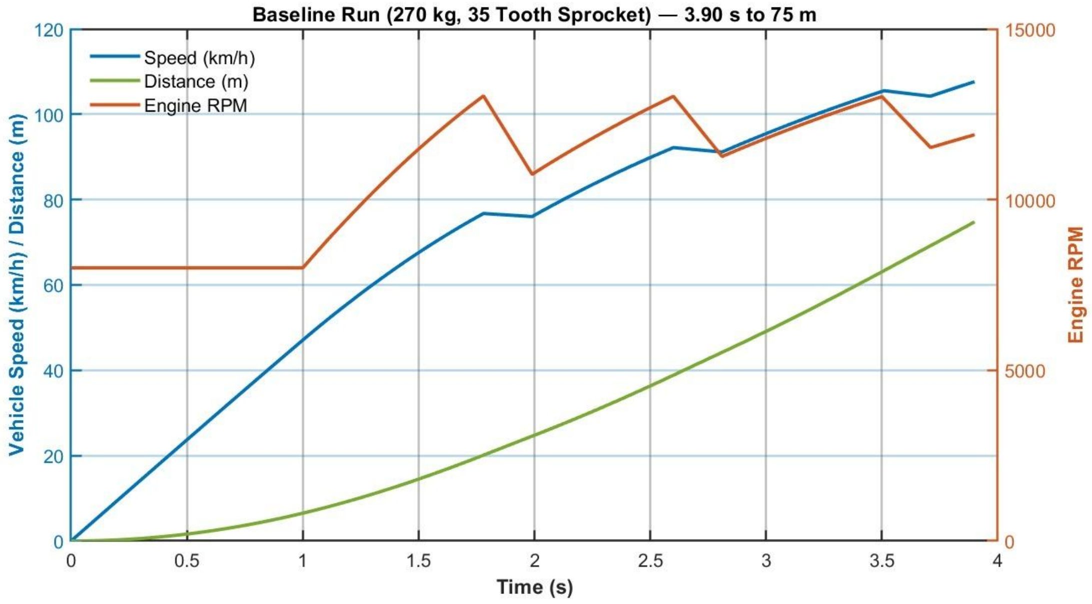
Project: Final Drive Ratio Optimization

- Wrote MATLAB scripts to sweep through vehicle parameters such as engine power, wheel slip torque, vehicle mass distribution, sprocket size, and shifting time.
- Want to optimize final drive ratio for a given vehicle and driver mass



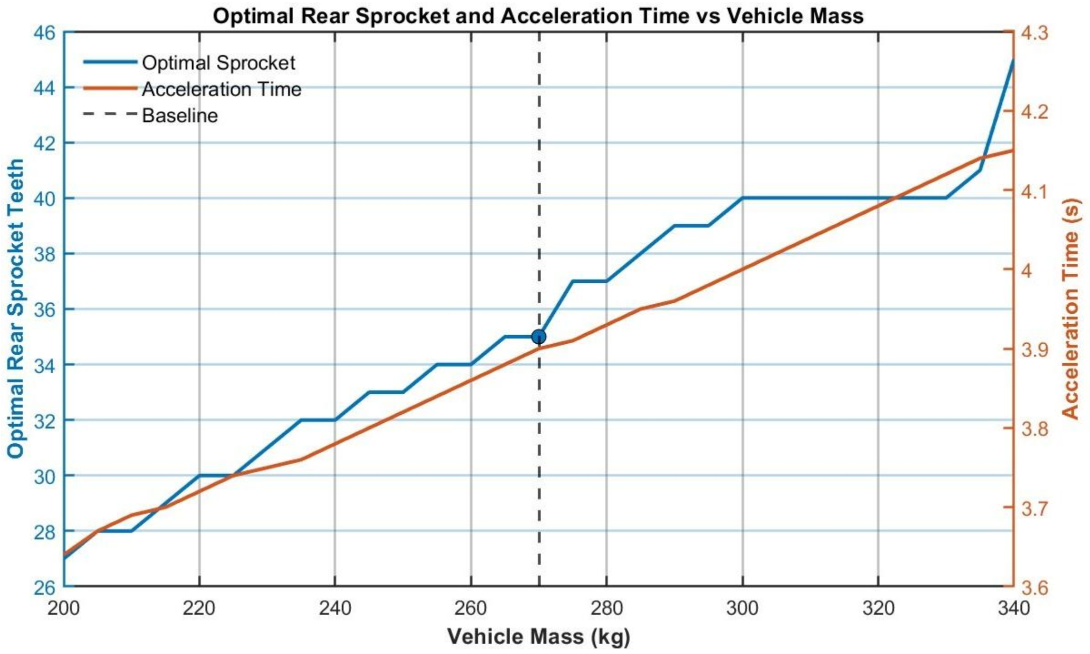
Baseline Acceleration Event Simulation

- Ran gear shift timing optimization simulation using estimated competition car and driver mass
- Optimal shift strategy starts in second gear and ends event in fifth gear

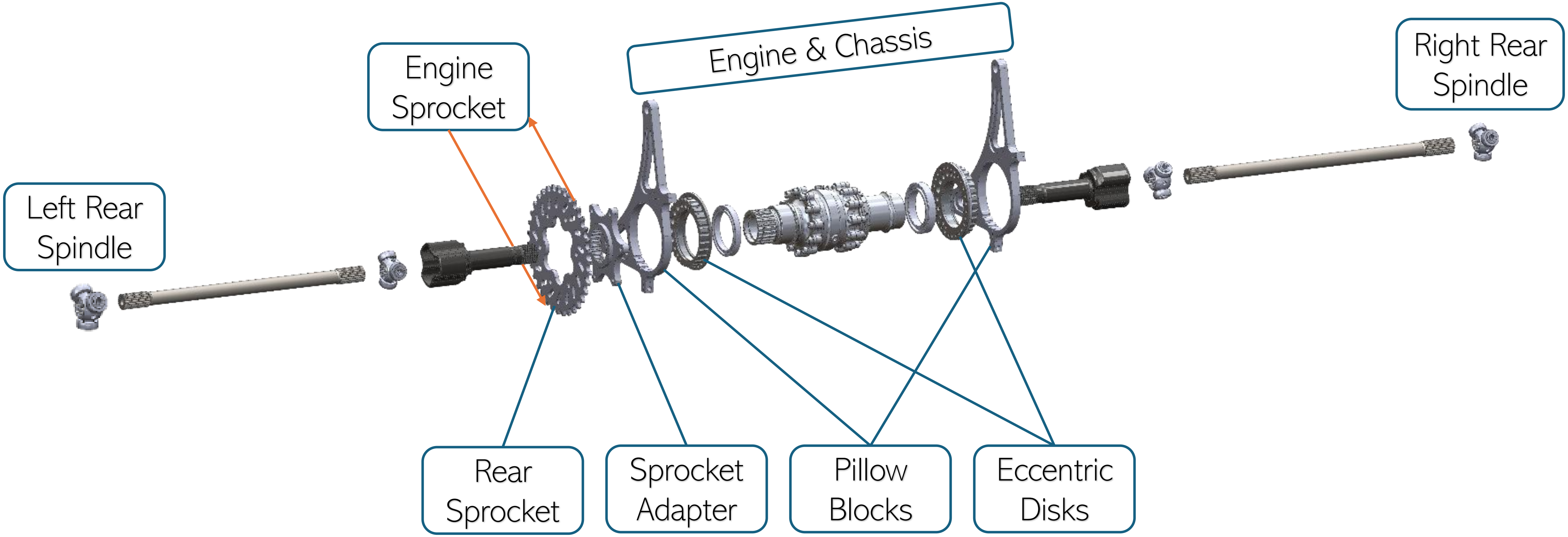


Acceleration Run Mass Sweep Simulation

- Swept through vehicle (driver) mass with different rear sprocket sizes
- Heavier vehicle benefits from shorter gearing (larger rear sprocket) and vice versa due
- Overall non-negligible change in 75-meter acceleration event time for changes in vehicle mass



Project: Drivetrain Structural Analysis



Drivetrain Structural Analysis

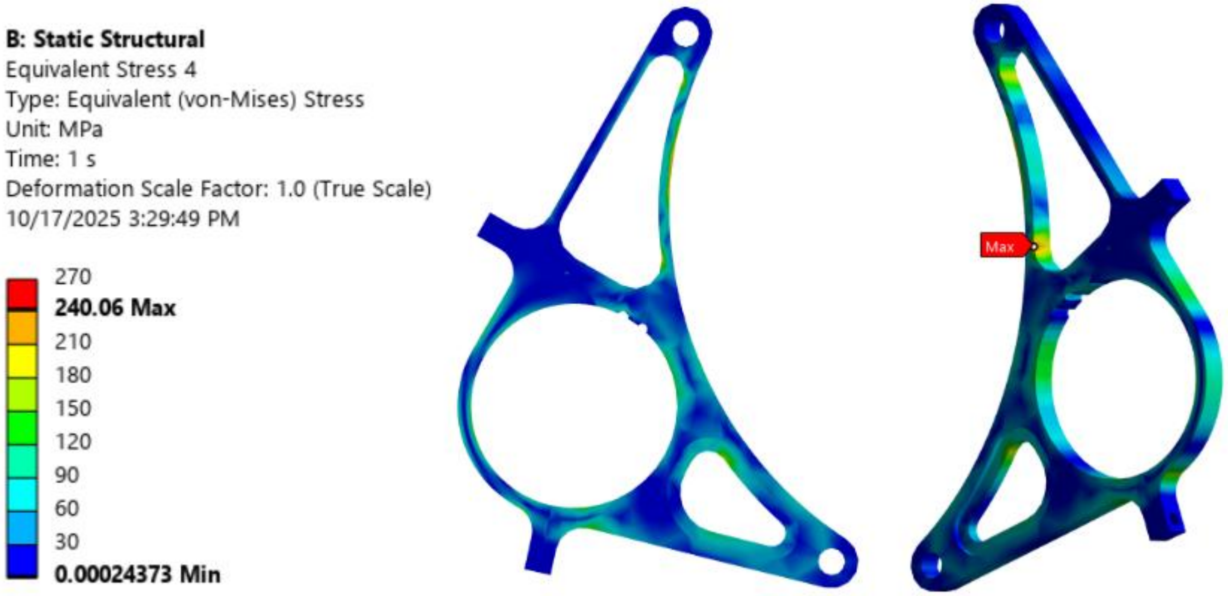
I am manufacturing the pillow blocks, eccentric bearing disks, rear sprocket, and sprocket adapter in house from Aluminum 7075 (ultimate-strength-limited)

Calculated maximum force vector on drivetrain components from engine torque curve and engine internal gear ratios

Currently collaborating with Engine lead on load case for a clutch drop that would exceed peak torque load on drivetrain

Left Pillow Block Structural Analysis

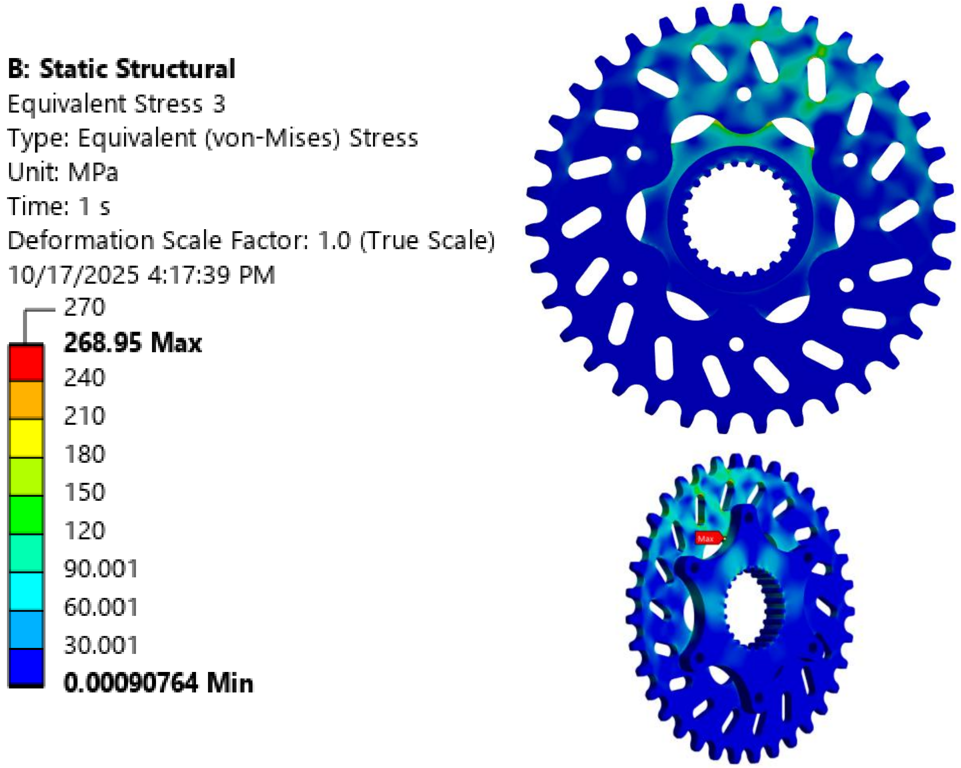
- FOS: Safety critical, replaceable
- Limited by 7075 ultimate strength
- Even stress distribution over supporting beams
- Good margin of safety (+0.12), weight reduced from last season



Loadcase	Solver	Analyst	Part	FOS (yield)	FOS (ultimate)	Yield Allowable	Ultimate Allowable	Ms_y (yield margin)	Ms_u (ultimate margin)
Peak Torque Chain Tension	ANSYS	B. Kovatchev	Left Pillow Block	1.25	2.0	235 MPa	270 MPa	**0.56**	**0.12**

Rear Sprocket & Adapter Structural Analysis

- FOS: Safety critical, replaceable
- Limited by 7075 ultimate strength
- Even stress distribution over teeth engaged with chain and connections between sprocket and adapter
- Excellent margin of safety (+0.004), weight minimized



Loadcase	Solver	Analyst	Part	FOS (yield)	FOS (ultimate)	Yield Allowable	Ultimate Allowable	Ms_y (yield margin)	Ms_u (ultimate margin)
Peak Torque Chain Tension	ANSYS	B. Kovatchev	Rear Sprocket	1.25	2.0	235 MPa	270 MPa	**0.39**	**0.004**

Project: Eccentric Bearing Disks

- Leading design process for rotating bearing disks to enable adjustable drivetrain position within pillow blocks and vehicle coordinates
- Taught drivetrain new members CAD and FEA basics in SolidWorks and ANSYS for eccentric disk design campaign
- Teaching new members manual and CNC machining principles to manufacture 2026 eccentric disks

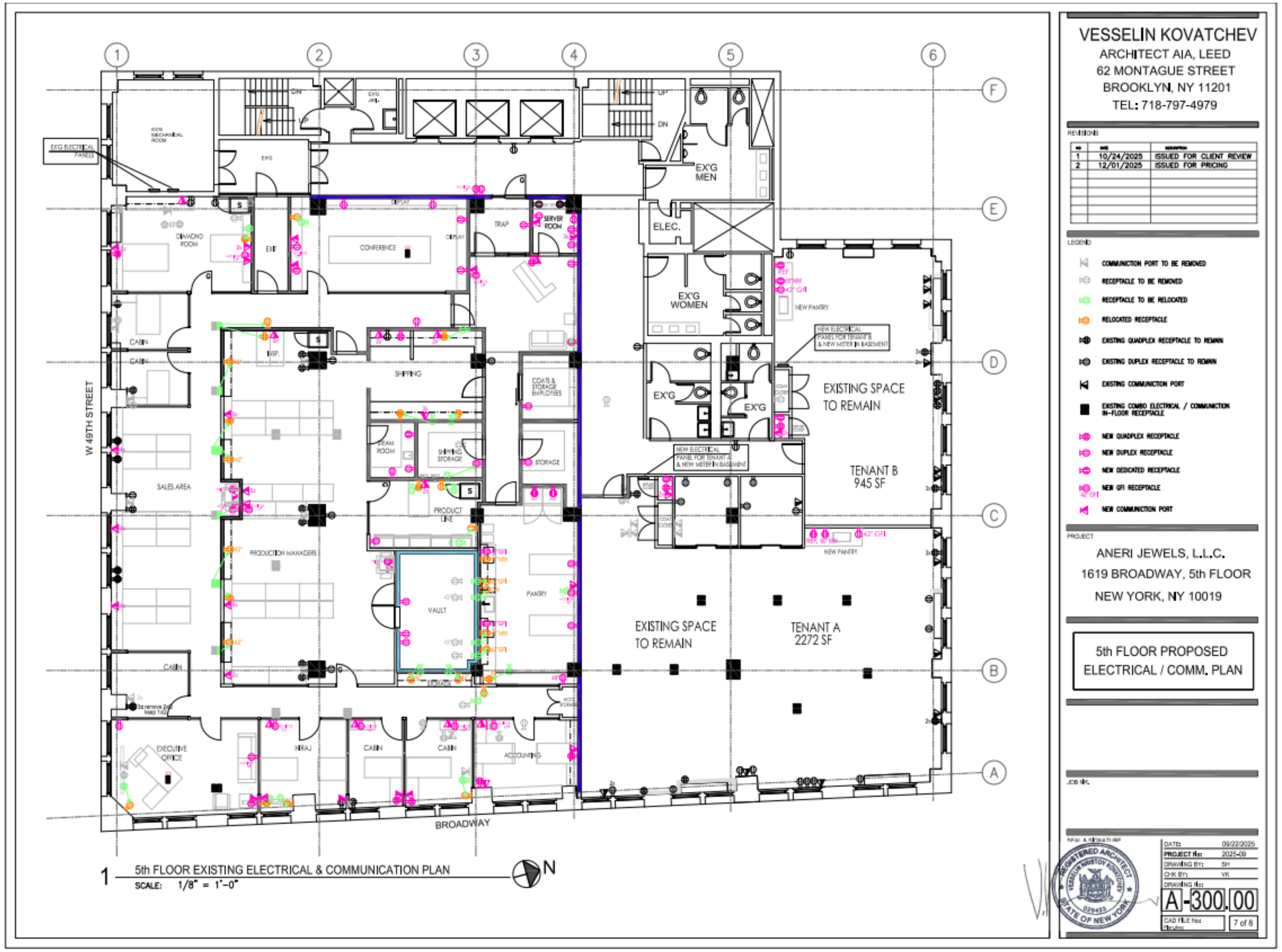
Personal Projects

Project: Electrical Load Plan & Analysis 1619 Broadway 5th Floor

I developed the electrical plan for power distribution and communications systems for a diamond and jewelry office. This included coordinating electrical layouts to support office operations while following applicable design standards.

I drafted the electrical and communications drawing package in AutoCAD that was submitted to the architect for design review and coordination. I then created a comprehensive load schedule and calculation in Excel, estimating maximum demand for major electrical loads such as servers, lighting, receptacles, communication ports, and HVAC-equipment. The completed drawings and load analysis were submitted to the electrical engineer and used as the basis for the project's electrical load letter.

Electrical & Communications Plan



- Architectural plan detailing electrical connections for power distribution
- Sent to architect for review and submittal

Load Schedule & Calculations

- Detailed maximum load calculation based on each device’s operating voltage and rated power, organized by room and system type.
- Consolidates lighting, receptacle, equipment, and specialty loads, providing a clear breakdown of maximum demand to support service load evaluation and engineering review.

⋮

44	**Load Per Section**					
45	**Total Maximum Load (W):**	**33769**				
46	Lobby | Total Max Load (W):	438				
47	Room Name		Type	Load (W)	Quantity	Total Load (W)
48	Elevator Lobby		Lighting J	4	57	228
49			Monitor	50	1	50
50	Mechnical Room		Lighting B	0	4	0
51			Condenser	0	1	0
52	Hallway 1		Lighting B	20	4	80
53	Hallway 2		Lighting B	20	4	80
54						
55						
56	Aneri | Total Max Load (W):	20767				
57	Room Name		Type	Load (W)	Quantity	Total Load (W)
58	Trap		Lighting D	18	5	90
59	Server Room		Lighting B	20	1	20
60			Server	0	1	0
61	Diamond Room		Lighting A	30	6	180
62			Computer	180	2	360
63			Monitor	50	1	50
64	Reception		Lighting J	4	55	220
65			Lighting J	4	38	152
66			Computer	180	1	180
67			Monitor	50	1	50
68	Hallway		Lighting G	15	17	255
69	Exit		Lighting B	20	2	40
70	Conference Room		Lighting D	18	24	432
71			Lighting F	18	2	36
72			Computer	180	1	180
73			Conference Room TV	100	1	100

⋮

Project: Cabin

Over the last two years, I have constructed a 12x12 ft cabin with a 6x12 ft deck area. Over the course of the project, I learned how to work with lumber, screws, saws, plywood, and roofing shingles. I learned how to pour concrete and mix cement and how to level out large surfaces. I learned about moisture control and insulation. I watched many videos online about similar projects and made a short YouTube series on building the foundation for the cabin.

The cabin has taken about 60 person-days to build, and I had a lot of help from friends and family. For their privacy I have only included pictures of the cabin and myself.

YouTube series first video: https://www.youtube.com/watch?v=p-m_M2N0r1k

Foundation

Foundation Beams

Concrete + 8x8 Posts

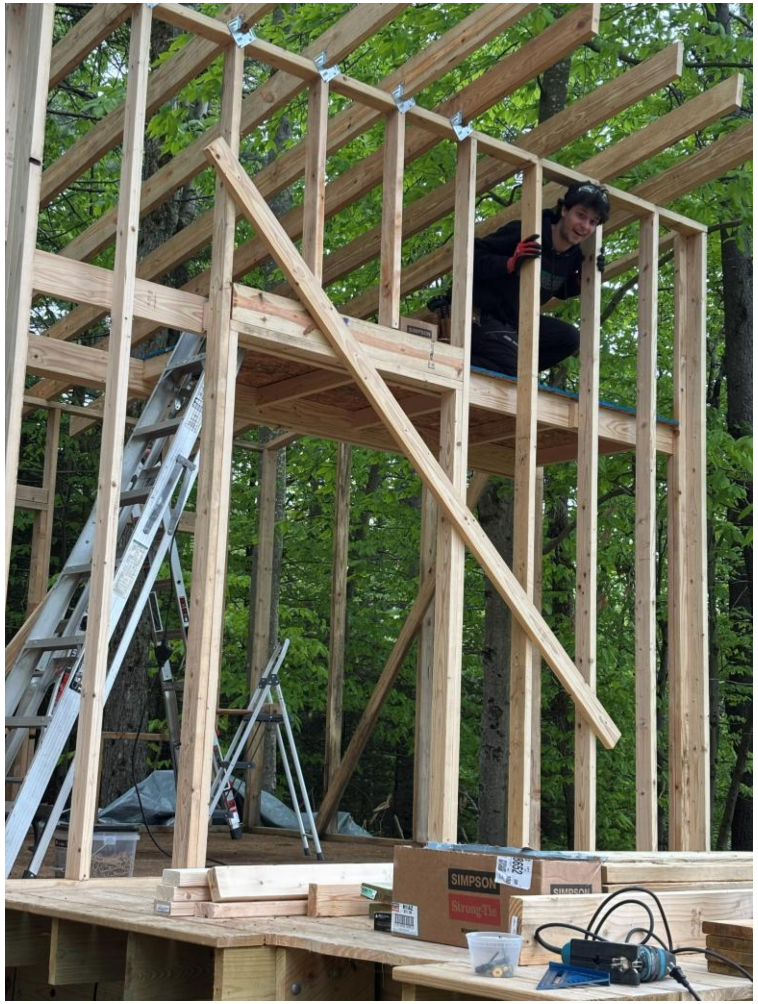
Floor and Deck Joists

Framing

First Wall with Windows

Stairs to Deck

Loft



Roofing

Upside Down T1-11 Siding

Plywood & Drip Edge

Roofing Felt & Asphalt Shingles

Finished Roof

Siding

Vapor Barrier

T1-11 Plywood Siding

Front View

Next Steps: Winter 2025-26

- Build windows
- Install electrical
- Finish interior
- Furnish with table, benches, chairs, bed in loft, and kitchenette

Thank You for Viewing

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